

1 Solving the PDE

We begin by writing down the problem explicitly, noting that the diffusion coefficient $D(t) = D(V(t))$ is time-dependent due to the voltage sweep:

$$\begin{cases} u_t = D(t)u_{xx} \\ u(0, t) = f(t) \\ u_x(L, t) = 0 \\ u(x, 0) = u(x, k\tilde{T}) \end{cases} \quad (1)$$

We can solve this equation by shifting the inhomogeneity from the boundary into the main differential equation. Let $u(x, t) = v(x, t) + f(t)$:

$$\begin{cases} v_t = D(t)v_{xx} - f'(t) \\ v(0, t) = 0 \\ v_x(L, t) = 0 \\ v(x, 0) + f(0) = v(x, k\tilde{T}) + f(k\tilde{T}) \end{cases} \quad (2)$$

We will approach this with the Spectral Method. We write:

$$v(x, t) = \sum_{n=1}^{\infty} T_n(t) \phi_n(x) \quad (3)$$

To match our boundary conditions, we conveniently choose the basis function $\phi_n(x) = \sin(\lambda_n x)$ with $\lambda_n = \frac{(2n-1)\pi}{2L}$. Putting this solution back into Eq. 2, we will solve for the $T_n(t)$ to obtain the full solution.

$$\frac{\partial}{\partial t} \left(\sum_{n=1}^{\infty} T_n(t) \sin(\lambda_n x) \right) = D(t) \frac{\partial^2}{\partial x^2} \left(\sum_{n=1}^{\infty} T_n(t) \sin(\lambda_n x) \right) - f'(t) \quad (4)$$

Next, we project $f'(t)$ onto our basis functions. Although $f'(t)$ is a constant in x , and is therefore even, we can perform an odd extension of our function and focus only on the interval of interest, $[0, L]$. This allows us to write:

$$f'(t) = f'(t) \sum_{n=1}^{\infty} c_n \sin(\lambda_n x) \quad (5)$$

where the coefficients are given by:

$$c_n = \frac{2}{L} \int_0^L \sin(\lambda_n x) dx \quad (6)$$

Substituting this back, we rewrite Eq. 4 as:

$$\frac{\partial}{\partial t} \left(\sum_{n=1}^{\infty} T_n(t) \sin(\lambda_n x) \right) = D(t) \frac{\partial^2}{\partial x^2} \left(\sum_{n=1}^{\infty} T_n(t) \sin(\lambda_n x) \right) - \sum_{n=1}^{\infty} f'(t) c_n \sin(\lambda_n x) \quad (7)$$

Multiplying both sides of the equation by $\sin(\lambda_m x)$ and integrating from 0 to L , we utilize the following orthogonality relationship:

$$\int_0^L \sin(\lambda_n x) \sin(\lambda_m x) dx = \frac{L}{2} \delta_{mn} \quad (8)$$

This trick gives us the following equation for each m :

$$T'_m(t) = -D(t) \lambda_m^2 T_m - f'(t) c_m \quad (9)$$

We solve this by using an integrating factor:

$$\frac{d}{dt} \left(T_m(t) \exp \left(\lambda_m^2 \int_0^t D(s) ds \right) \right) = -f'(t) c_m \exp \left(\lambda_m^2 \int_0^t D(s) ds \right) \quad (10)$$

We integrate over each individual time step $[t_{i-1}, t_i]$ and define $\Delta t_i = t_i - t_{i-1}$. Assuming the time step is small enough that $D(t) \approx D(t_i)$ is roughly constant over this interval, we obtain:

$$T_m(t_i) = T_m(t_{i-1}) e^{-D(t_i) \lambda_m^2 \Delta t_i} - c_m \int_{t_{i-1}}^{t_i} f'(\tau) e^{-D(t_i) \lambda_m^2 (t_i - \tau)} d\tau \quad (11)$$

Assuming Δt_i is sufficiently small, the exponential inside the integral is approximately one for $\tau \in [t_{i-1}, t_i]$. This yields:

$$\begin{aligned} T_m(t_i) &\approx T_m(t_{i-1}) e^{-D(t_i) \lambda_m^2 \Delta t_i} - c_m \int_{t_{i-1}}^{t_i} f'(\tau) d\tau \\ &= T_m(t_{i-1}) e^{-D(t_i) \lambda_m^2 \Delta t_i} + c_m [f(t_{i-1}) - f(t_i)] \end{aligned}$$

Evaluating this iteratively for $i = 1, 2, \dots$, we identify a pattern and can explicitly write:

$$T_m(t_N) = T_m(t_0) \left(\prod_{i=1}^N e^{-D(t_i) \lambda_m^2 \Delta t_i} \right) + \sum_{i=1}^N \left(c_m [f(t_{i-1}) - f(t_i)] \prod_{j=i+1}^N e^{-D(t_j) \lambda_m^2 \Delta t_j} \right) \quad (12)$$

Combining this with the periodic condition $T_m(0) = T_m(k\tilde{T})$, we can rewrite the expression to solve for the starting condition:

$$T_m(0) = \frac{\sum_{i=1}^N \left(c_m [f(t_{i-1}) - f(t_i)] \prod_{j=i+1}^N e^{-D(t_j) \lambda_m^2 \Delta t_j} \right)}{1 - \left(\prod_{i=1}^N e^{-D(t_i) \lambda_m^2 \Delta t_i} \right)} \quad (13)$$

With the initial periodic state $T_m(0)$, we can evaluate $T_m(t_i)$ for all time steps. Substituting these coefficients yields the solution to the problem for $v(x, t)$. Finally, by adding the boundary function $f(t)$ back, we recover the complete theoretical solution to the original partial differential equation:

$$u(x, t) = f(t) + \sum_{m=1}^{\infty} T_m(t) \sin(\lambda_m x) \quad (14)$$

2 Numerical Implementation

To map the mathematical solution to experimental Cyclic Voltammetry (CV) data, the model was implemented using Python. We utilize the JAX library for Just-In-Time (JIT) compilation and automatic differentiation to solve the inverse problem.

2.1 Modeling the Boundary Condition and Density of States

The time-dependent boundary condition $f(t)$ represents the equilibrium surface concentration driven by $V(t)$. This is determined by integrating the Density of States (DOS) up to the current voltage:

$$f(V) = \int_{-\infty}^V \text{DOS}(v) dv \quad (15)$$

To capture the complex redox behavior of the system, we model the true DOS as a superposition of P independent Gaussian peaks. The exact analytical integral of a Gaussian yields an Error Function (erf).

However, evaluating the Error Function iteratively is computationally expensive. To bypass this, we approximate the integrated Gaussians using a sum of logistic functions. We parameterize the boundary condition as:

$$f(V) = \sum_{p=1}^P \frac{w_p}{1 + \exp[-\gamma_p(V - V_{c,p})]} \quad (16)$$

where for each constituent peak p , w_p is the weight (capacity), $V_{c,p}$ is the critical peak voltage, and γ_p dictates the peak sharpness. This formulation allows JAX to rapidly compute both $f(t)$ and the DOS analytically without relying on expensive numerical integration.

2.2 Forward Model and Time-Stepping

To model the time-dependent diffusion coefficient, we use a phenomenological Gaussian distribution around a central voltage V_c , with asymmetric decay rates β_L and β_R for the anodic and cathodic sweeps:

$$D(V) = D_0 \exp[\beta_{L,R}(V - V_c)^2]. \quad (17)$$

The periodic initial condition is computed directly using Eq. 13. This recursion is calculated rapidly using `jax.lax.scan`, which accumulates the exponential decay and forcing matrices.

The total number of particles $N(t)$ in the film is found by integrating the solution $u(x, t)$ over the spatial domain $[0, L]$. The pure Faradaic current is then given by $I = \frac{dN}{dt}$, evaluated numerically as a finite difference. To accurately match experimental data, we add a constant baseline offset (I_{offset}) and background exponential tails simulating Tafel-like kinetics at the voltage boundaries:

$$I_{\text{sim}}(t) = \frac{\Delta N}{\Delta t} + I_{\text{offset}} + a_R e^{k_R(V - V_{\text{max}})} - a_L e^{-k_L(V - V_{\text{min}})} \quad (18)$$

2.3 Optimization Strategy

To extract the physical parameters, we frame the task as an optimization problem. The objective function minimizes the weighted mean squared error (MSE) between the simulated current I_{sim} and the experimental current I_{exp} .

To ensure the optimizer captures the sharp Faradaic peaks rather than merely fitting the baseline, the loss function is weighted using the second derivative of the smoothed experimental current:

$$W \propto 1 + \frac{\left| \frac{d^2 I_{\text{exp}}}{dt^2} \right|}{\max \left| \frac{d^2 I_{\text{exp}}}{dt^2} \right|} \quad (19)$$

Because the parameter space is complex, finding a global minimum requires a structured approach. We minimize the loss function using the L-BFGS-B algorithm, leveraging exact gradients computed via JAX’s reverse-mode automatic differentiation (`jax.value_and_grad`). To avoid local minima, the optimization is executed in four targeted stages:

1. **Pure Flat Baseline:** Parameters are frozen except for the constant current offset to align the simulation with the data.
2. **Background Tails:** The exponential parameters ($a_{L,R}, k_{L,R}$) are optimized to fit the edges of the voltage sweep.
3. **Anchor Peaks (Constant D):** The diffusion coefficient is held constant while the heights and positions of the Gaussian peaks defining the density of states (DOS) are optimized.
4. **Full Non-Linear Polish:** All bounds are relaxed. The full set of parameters, including the asymmetric voltage-dependent diffusion parameters β_L and β_R , are optimized simultaneously to achieve the final fit.

Upon convergence, this routine yields the voltage-dependent diffusion $D(V)$ and the Density of States for the experimental chemical system.